2. Model

Causal structure, full specification, priors and Stan code

The model estimates both links of the paradigm jointly, morphology → performance and performance → survival, while correcting survival for imperfect detection. The choices behind the data inputs are explained on the Data page. A PDF version of this page is also available.

1 Causal diagram

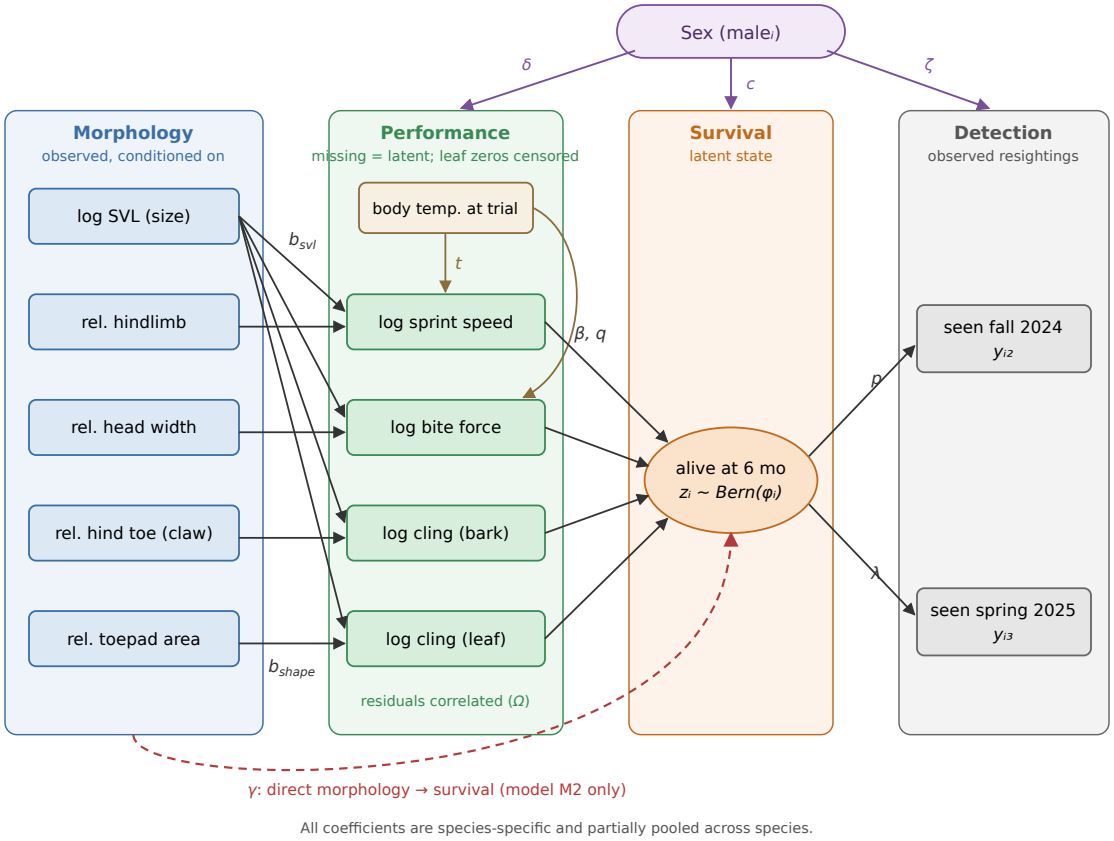


Figure 1: Causal structure of the model. Solid arrows are present in both models. The dashed arrow (direct effects γ) is present only in M2. Sex affects performance (δ), survival (c) and detection (ζ). Body temperature at the trial affects sprint and bite (t).

2 Transformations

All continuous variables except body temperature are log-transformed. All, including temperature, are then standardised within species, using that species' observed sample mean and SD:

$$\mathrm{std}_s(v) = \frac{v - \bar{v}_s}{\mathrm{sd}_s(v)}.$$

Coefficients are then in within-species SD units, and so comparable among species and traits. Survival coefficients become standardised selection gradients (on the logit scale), and one set of generic priors can be used. The constants $\bar{v}_s$ and $\mathrm{sd}_s(v)$ are fixed data, so the simulator used for calibration applies the identical transformation (`prepare_data()` in `R/prepare-data.R`).

3 Notation

Symbol	Meaning
$i = 1, \ldots, N$	individual
$s = s[i] \in \{\text{caro}, \text{cris}, \text{sag}\}$	species of individual i
$j \in \{\text{spr}, \text{bite}, \text{bark}, \text{leaf}\}$	performance trait
$k \in \{\text{svl}, \text{hl}, \text{hw}, \text{toe}, \text{pad}\}$	morphological predictor
$k(j)$	shape trait linked to j: $k(\text{spr}) = \text{hl}$, $k(\text{bite}) = \text{hw}$, $k(\text{bark}) = \text{toe}$, $k(\text{leaf}) = \text{pad}$
$m_i \in \{0, 1\}$	male indicator (female = 0)
x_{ik}	standardised morphological predictor (data)
w_{ij}	standardised body temperature at the trial for j (data; 0 for bark and leaf)
u_{ij}	standardised log performance (data, or latent if missing)
$z_i \in \{0, 1\}$	alive at the fall 2024 census (latent)
$y_{i2}, y_{i3} \in \{0, 1\}$	seen in fall 2024 / spring 2025

4 Performance submodel

Each lizard's vector of the four performance traits is multivariate normal:

$$\mathbf{u}_i \sim \mathrm{MVN}\left(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{s[i]}\right), \qquad \boldsymbol{\Sigma}_s = \mathrm{diag}\,(\boldsymbol{\sigma}_s)\,\boldsymbol{\Omega}\,\mathrm{diag}\,(\boldsymbol{\sigma}_s),$$

with a mean that depends on size, the linked shape trait, sex and body temperature:

$$\mu_{ij} = \alpha_{sj} + \delta_{sj}\, m_i + b^{\text{svl}}_{sj}\, x_{i,\,\text{svl}} + b^{\text{shape}}_{sj}\, x_{i,k(j)} + t_{sj}\, w_{ij}.$$

- α_{sj} is the expected performance of a female of average size and shape, tested at the species' mean temperature.
- t_{sj} is the effect of body temperature during the trial. It is estimated for sprint and bite only, and fixed at 0 for both clinging traits. Temperature is a condition of the trial, not a trait of the lizard. Survival therefore depends on temperature-adjusted performance, $\tilde{u}_{ij} = u_{ij} - t_{sj}\, w_{ij}$: the lizard's expected performance at its species' mean test temperature.

- δ_{sj} is the male–female difference in performance not explained by morphology.
- b_{sj}^{svl} and b_{sj}^{shape} are the size and shape effects on performance (the first link of the paradigm).
- $\boldsymbol{\Omega}$ is a residual correlation matrix shared across species. It captures lizards that perform well on all tasks for reasons beyond the measured morphology, such as condition or motivation.
- Missing u_{ij} are parameters drawn from the same distribution, so lizards with missing performance still contribute their survival information. This assumes values are missing at random given morphology, sex and the other performance traits.
- Leaf-clinging zeros are left-censored. For those lizards, $u_{i,\,\text{leaf}}$ is also a parameter, but it is constrained to lie below the detection limit,

$$u_{i,\,\text{leaf}} < c_s = \text{std}_s(\log 0.01),$$

  and follows the same multivariate normal distribution. They therefore count as the weakest leaf clingers, rather than being dropped or given an arbitrary value. The standardisation constants for leaf clinging are computed from the non-zero values.

5 Survival and detection submodel

The probability of surviving the first six months depends on temperature-adjusted performance:

$$z_i \sim \text{Bernoulli}\ (\phi_i),$$

$$\text{logit}\ (\phi_i) = a_s + c_s\, m_i + \sum_j \beta_{sj}\, \tilde{u}_{ij} + \sum_j q_{sj}\, \tilde{u}_{ij}^2 \underbrace{\left[+ \sum_k \gamma_{sk}\, x_{ik} \right]}_{\text{M2 only}}.$$

- a_s is the logit 6-month survival of an average female.
- c_s is the sex difference in survival.
- β_{sj} is linear selection on performance j, and q_{sj} is quadratic selection. $q < 0$ is stabilising and $q > 0$ disruptive. The conventional quadratic gradient is $2q_{sj}$.
- γ_{sk} is the direct effect of morphology on survival. It is fixed at 0 in M1. Including $x_{i,\,\text{svl}}$ here guards against attributing size-related survival to performance.
- Correlational selection ($\tilde{u}_{ij}\tilde{u}_{ij'}$ terms) is not included.

Resighting depends on whether the lizard is alive:

$$y_{i2} \sim \text{Bernoulli}\ (z_i\, p_i), \qquad \text{logit}\ (p_i) = \pi_s + \zeta_p\, m_i,$$

$$y_{i3} \sim \text{Bernoulli}\ (z_i\, \lambda_i), \qquad \text{logit}\ (\lambda_i) = \ell_s + \zeta_\lambda\, m_i.$$

- p_i is the probability of detection in fall given the lizard is alive.
- λ_i is the probability of being seen in spring 2025 given the lizard was alive in fall. It combines fall-to-spring survival and spring detection, which three occasions cannot separate. It is a nuisance parameter.
- The sex effects on detection, ζ_p and ζ_λ, are shared across species. Males display and may be more conspicuous.

This is a Cormack–Jolly–Seber model for three occasions. Summing over the latent z_i gives the likelihood of each history:

History (y_{i2}, y_{i3})	Probability
1 1	$\phi_i \, p_i \, \lambda_i$
0 1	$\phi_i \, (1 - p_i) \, \lambda_i$
1 0	$\phi_i \, p_i \, (1 - \lambda_i)$
0 0	$(1 - \phi_i) + \phi_i \, (1 - p_i)(1 - \lambda_i)$

Lizards seen in spring 2025 but missed in fall (history 01) identify p. Selection on 6-month survival (ϕ) is then estimated free of detection bias. This assumes detection does not depend on performance or morphology beyond sex.

6 Derived quantities

For morphological predictor k in species s, evaluated at the species mean, where $\tilde{u} = 0$ so the quadratic terms vanish:

$$\text{indirect}_{sk} = \sum_j b_{sjk} \, \beta_{sj}, \qquad \text{direct}_{sk} = \gamma_{sk}, \qquad \text{total}_{sk} = \text{indirect}_{sk} + \text{direct}_{sk}.$$

Here b_{sjk} is b_{sj}^{svl} for $k = \text{svl}$, b_{sj}^{shape} when $k = k(j)$, and 0 otherwise. Size is the only predictor with indirect paths through all four performance traits. Support for full mediation is assessed in three ways: from the posteriors of γ_{sk}, by comparing M1 and M2 with PSIS-LOO, and by checking in simulation (Validation) whether the design can detect $\gamma \neq 0$.

7 Priors

7.1 Hierarchical structure

Every parameter labelled hierarchical below varies by species and is partially pooled:

$$\theta_s = \mu_\theta + \tau_\theta \, \tilde{\theta}_s, \qquad \tilde{\theta}_s \sim \mathcal{N}(0, 1),$$

with separate μ_θ and τ_θ for each performance trait j or predictor k. This is the non-centered form used in Stan. With only three species, τ_θ is weakly identified, so its priors are deliberately informative.

7.2 Prior distributions

All scales are standardised traits or logits. $\mathcal{N}^+$ is the half-normal distribution.

Parameter	Pooling	Prior
Performance		
α_{sj}	none	$\mathcal{N}(0, 0.5)$
δ_{sj}	hierarchical	$\mu \sim \mathcal{N}(0, 1)$, $\tau \sim \mathcal{N}^+(0, 0.5)$
b_{sj}^{svl}	hierarchical	$\mu \sim \mathcal{N}(0, 0.5)$, $\tau \sim \mathcal{N}^+(0, 0.25)$

Parameter	Pooling	Prior
b_{sj}^{shape}	hierarchical	$\mu \sim \mathcal{N}(0, 0.5)$, $\tau \sim \mathcal{N}^+(0, 0.25)$
t_{sj} (sprint, bite)	hierarchical	$\mu \sim \mathcal{N}(0, 0.5)$, $\tau \sim \mathcal{N}^+(0, 0.25)$
σ_{sj}	none	$\mathcal{N}^+(0, 1)$
$\mathbf{\Omega}$	shared	$\text{LKJ}\ (2)$
Survival		
a_s	hierarchical	$\mu \sim \mathcal{N}(-1, 1)$, $\tau \sim \mathcal{N}^+(0, 0.5)$
c_s	hierarchical	$\mu \sim \mathcal{N}(0, 1)$, $\tau \sim \mathcal{N}^+(0, 0.5)$
β_{sj}	hierarchical	$\mu \sim \mathcal{N}(0, 0.5)$, $\tau \sim \mathcal{N}^+(0, 0.25)$
q_{sj}	hierarchical	$\mu \sim \mathcal{N}(0, 0.25)$, $\tau \sim \mathcal{N}^+(0, 0.1)$
γ_{sk} (M2)	hierarchical	$\mu \sim \mathcal{N}(0, 0.5)$, $\tau \sim \mathcal{N}^+(0, 0.25)$
Detection		
π_s	hierarchical	$\mu \sim \mathcal{N}(0, 1.5)$, $\tau \sim \mathcal{N}^+(0, 1)$
ℓ_s	hierarchical	$\mu \sim \mathcal{N}(0, 1.5)$, $\tau \sim \mathcal{N}^+(0, 1)$
ζ_p, ζ_λ	shared	$\mathcal{N}(0, 1)$

The reasoning behind these priors:

- **Performance intercepts $\boldsymbol{\alpha_{sj}}$** are near 0 by construction, because traits are centred within species. They are offset only by the sex term, so pooling them adds nothing. A first version used $\mathcal{N}(0, 1)$, but prior predictive simulation showed it allowed implausible shifts in performance, for example leaf censoring rates above 30% when about 3% is observed. It was tightened to $\mathcal{N}(0, 0.5)$.
- **Performance slopes** of $\mathcal{N}(0, 0.5)$ put 95% of the mass below about 1 SD of performance per SD of morphology.
- **Residual SDs σ_{sj}** must lie below 1, because performance is standardised.
- **Baseline survival** $\mu_a \sim \mathcal{N}(-1, 1)$ centres 6-month survival near 27%. That is a little above the naive 20%, as expected once detection is accounted for.
- **Selection gradients** of $\mathcal{N}(0, 0.5)$ allow odds of survival to change up to about 2.7-fold per SD of performance (2 prior SDs). The quadratic terms are tighter.
- **Detection intercepts** of $\mathcal{N}(0, 1.5)$ are close to uniform on the probability scale.

Prior predictive checks of these priors are on the Validation page.

8 Assumptions and limitations

- **Performance and morphology are measured without error.** Sprint timing resolution violates this mildly.
- **Missing data.** Performance is missing at random given the model. The 17 lizards with missing toepad area or hind toe length are excluded. Only one of them survived to 6 months.
- **Detection** depends on species and sex only. Movement off the study area is indistinguishable from death, so ϕ is *apparent* survival.
- **Shared residual correlation.** The correlation among performance traits, $\boldsymbol{\Omega}$, is the same for all species.
- **Temperature.** Its effect on performance is linear, and it is unmeasured for both clinging trials.
- **Claw size.** Relative hind toe length is a stand-in for claw size.
- **No correlational selection.** It could be added as an extension.
- **Three species.** Among-species variances are estimated mostly from their priors.

9 Stan code

> **i** In progress
>
> The Stan implementation will be added here. It will read the same priors as the simulator (`R/priors.R`) and use one flag to switch between M1 and M2.